

Assessment of Children Early Language Skills

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Brief Recap

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Project:

- Create a language skill assessment for children aged 0-5, with different sets of questions designed for each age group (0-1, 1-2, 2-3, 3-4, and 4-5 years)
- Client's goal: build a scoring system to track children's language growth.

Data collection (expected):

- Questionnaire for different age groups of children. Questionnaires will be collected every 6 month, followed up for 4 years.

Our Objectives:

- Using R to generate the synthetic data.
- Using proportional odds model to predict a child's response at different age groups based on his/her previous answers.
- Compare the model's predictions to the “true responses” from the synthetic data to evaluate prediction accuracy.

The Proportional Odds Model

The proportional odds model is a class of generalized linear model used for modelling the dependent of an ordinal response on discrete or continuous covariates.

$$\text{logit}(\gamma_j) = \alpha_j - \beta_j^T x,$$

α : the intercept that depends on level j , moves the cutoff point for each category.

β : the slope that is assumed to be constant across all thresholds.

Data stimulation

Age group	Questions			
		Q1	Q2	Q3
	Year 0	a11	a12	a13
	Year 0.5			
	Year 1			
	...			

One model per question per time window

Data stimulation

Questions

Age group

	Q1	Q2	Q3
Year 0	a11	a12	a13
Year 0.5			
Year 1			
...			

Algorithm 1 Pseudo Code for Sampling Y

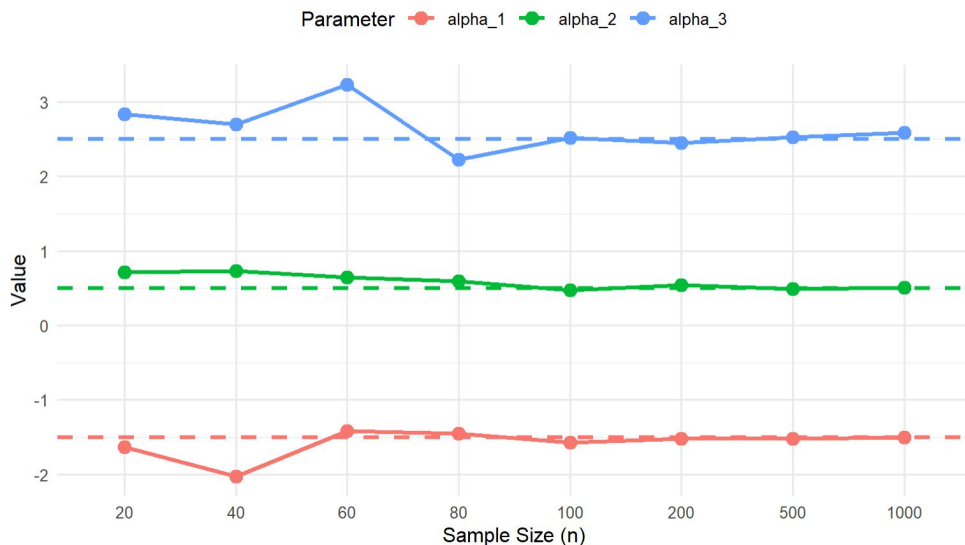
```
1:  $Y \leftarrow \text{array}(0, \text{dim} = (n, h, k))$ 
2: for  $i = 1$  to  $n$  do
3:   for  $\ell = 1$  to  $h$  do
4:     for  $j = 1$  to  $k$  do
5:       if  $\ell = 1$  then
6:          $X \leftarrow \text{array}(0, \text{dim} = (k))$ 
7:       else
8:          $X \leftarrow Y[i, \ell - 1, :]$ 
9:       end if
10:       $\text{logodds} \leftarrow \alpha - \text{crossprod}(X, \beta)$ 
11:       $P_{ij} \leftarrow f(\text{logodds})$ 
12:       $y_{ij} \leftarrow \text{sample categorical with } P_{ij}$ 
13:       $Y[i, \ell, j] \leftarrow y_{ij}$ 
14:     end for
15:   end for
16: end for
```

▷ pseudo

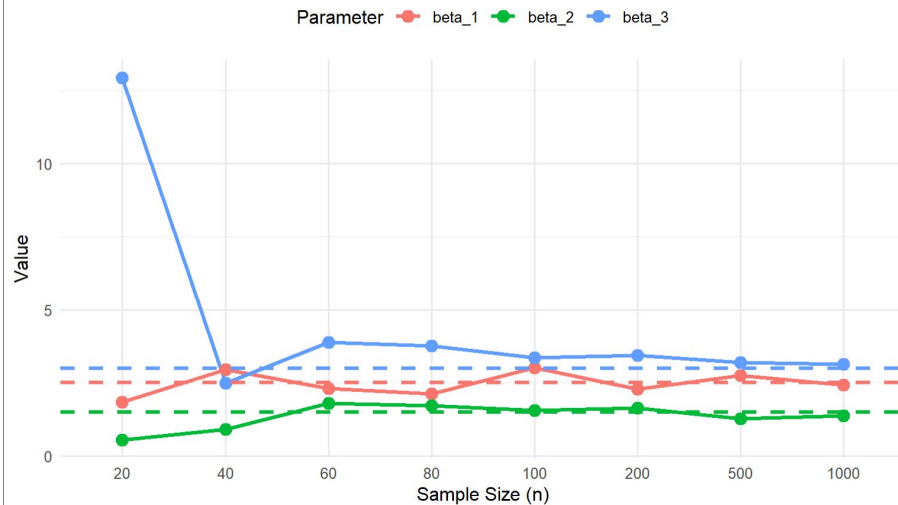
▷ pseudo

Simple version: 1000 kids, 3 questions, 3 age groups
 $Y=0,1,2,3$
 $\alpha \leftarrow c(-1.5, 0.5, 2.5)$
 $\beta \leftarrow c(2.5, 1.5, 3)$

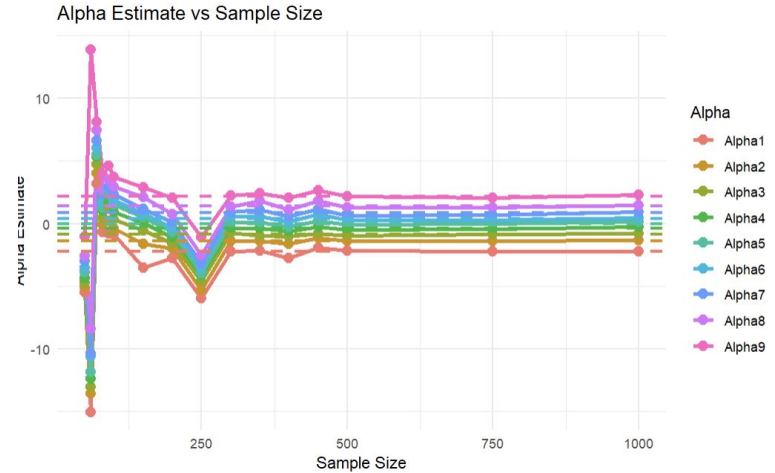
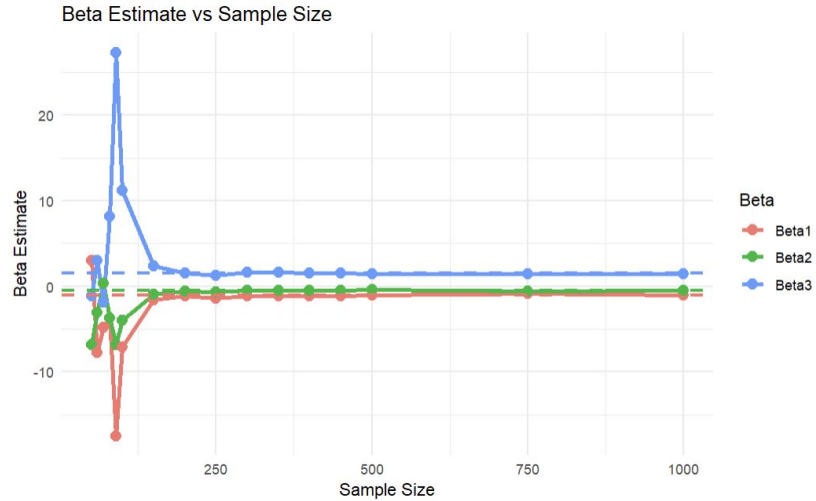
Alpha Parameters: True vs. Estimated Values



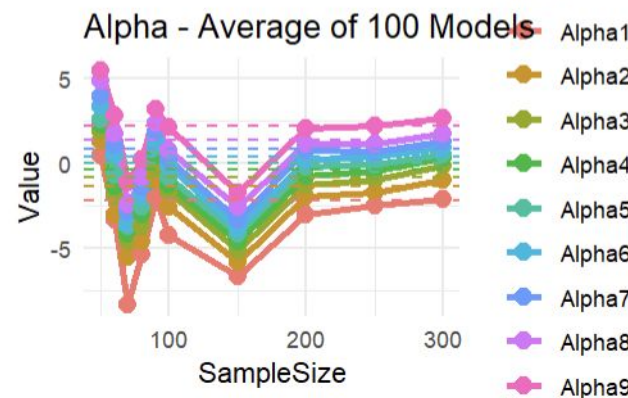
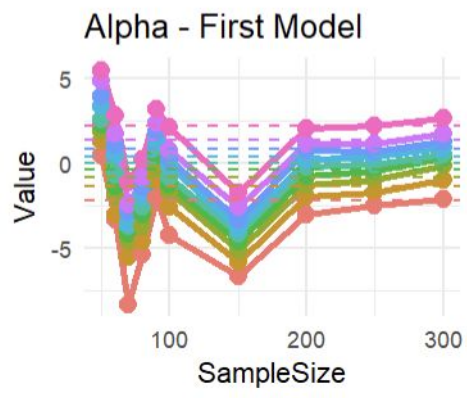
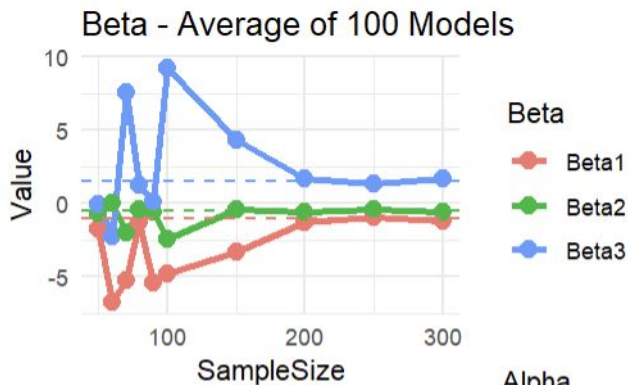
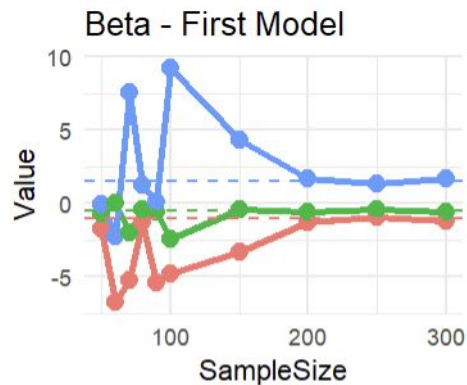
Beta Parameters: True vs. Estimated Values



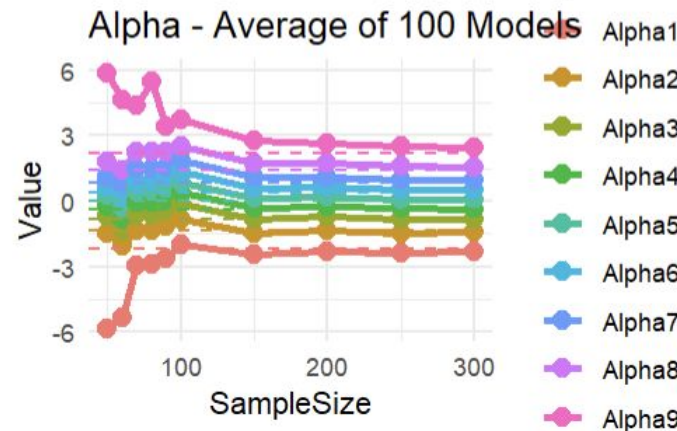
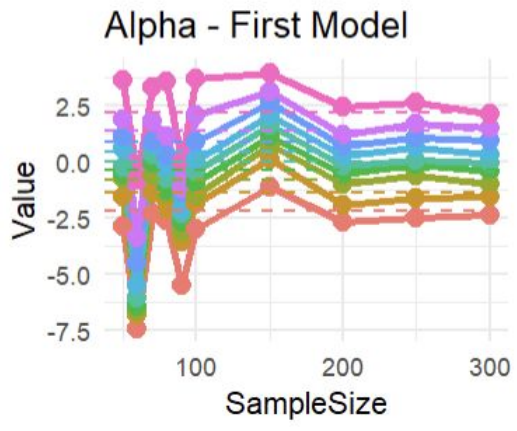
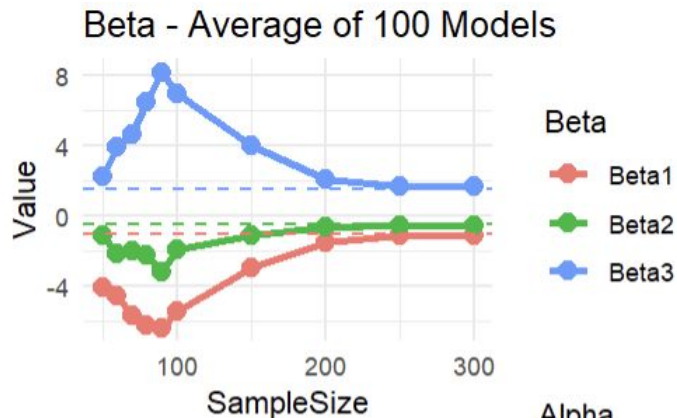
Converge point



Randomness in modeling algorithm (Same data)

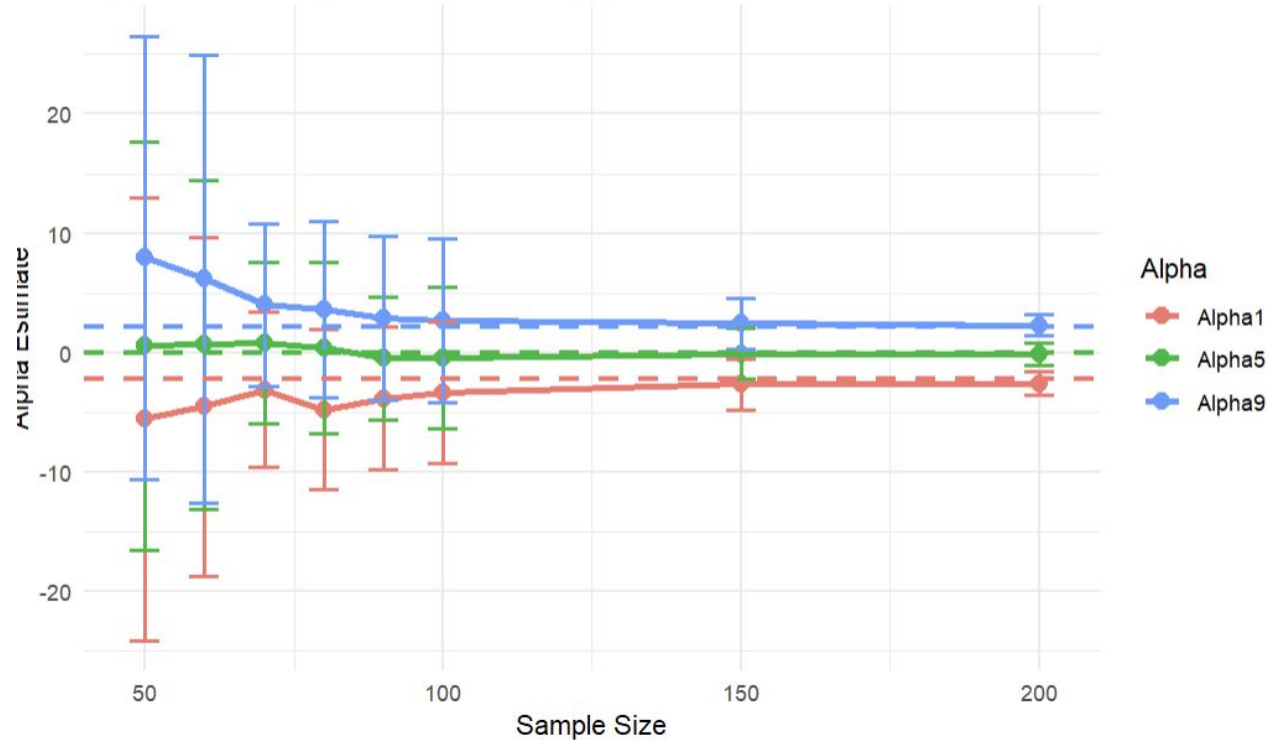


Randomness in modeling algorithm (Different data)



Spread of estimate value

Alpha Estimates (First, Middle, Last) with SD across Models



Thank You